

1 Newcomb 1898

Newcomb 1898, Page 9

I. BASIS OF THE TABLES.

The elements, masses, and other data on which these tables are based are those derived in the author's work entitled *The elements of the four inner Planets and the fundamental Constants of Astronomy*, forming a supplement to the *American Ephemeris and Nautical Almanac* for the year 1897. An ulterior correction of $-0''.33 + 0''.30 T$ (1850) has been applied to the undisturbed mean longitude to balance the change in terms of long period. The variable quantities have been reduced to the epoch

1900, Jan. 0, Greenwich Mean Noon,

as the fundamental epoch of the tables. The time from this epoch, reckoned in terms of the Julian century, or 36525 days as the unit, is represented by the symbol T .

The following are the expressions for the principal elements of the Earth's motion around the Sun, or of the Sun's apparent motion around the Earth, as derived from the data in question.

The Sun's geometric mean longitude, freed from aberration ;

$$L = 279^\circ 41' 48''.04 + 129\,602\,768''.13 T + 1''.089 T^2$$

Notes

Is T in increments of $1/36525$, or can it be fractional?

Or, with the base angle in decimal degrees instead of degrees-minutes-seconds,

$$L = 279.6966\bar{7}^\circ + 8\,640\,184^s.542 T + 0^s.0726 T^2$$

using $279 + 41/60 + 48.04/3600 = 279 + 62701/90000 = 279.6966\bar{7}$,

$$129\,602\,768''.13 \cdot \frac{1^s}{15''} = 8\,640\,184^s.542^s$$

and

$$1''.089 \cdot \frac{1^s}{15''} = 0.0726^s$$

In terms of hour-angle (unconventional in this form because not an astronomical time, but useful for comparison to τ)

$$L = 18^h 38^m 47^s.2 + \dots$$

using

$$(279 + 41/60 + 48/3600)^\circ \cdot (\text{hr}/15^\circ) = (18 + 2909/4500)^h,$$

$$(2909/4500)^h \cdot (60 \text{ min/hr}) = (38\frac{59}{75})^m, \text{ and}$$

$$(59/75)^m \cdot (60 \text{ s/min}) = (47\frac{1}{5})^s.$$

The Sun's mean sidereal motion in a Julian year;

$$n = 1\ 295\ 977''.4320 - 0''.000403\ T$$

The longitude of the solar perigee, also freed from aberration ;

$$\pi = 281^\circ\ 13'\ 15''.0 + 6189''.03\ T + 1''.63\ T^2 + 0''.012\ T^3$$

Notes

Or

$$\pi = 281.2208\overline{3}^\circ + \dots$$

$$\pi = 18^h 44^m 53^s.0 + \dots$$

using

$$281 + 13/60 + 15/3600 = 281\frac{53}{240} = 281.2208\overline{3},$$

$$(281 + 13/60 + 15/3600)/15 = (18 + 2693/3600)^h,$$

$$(2693/3600) \cdot 60 = (44 + 53/60)^m, \text{ and}$$

$$(53/60) \cdot 60 = 53^m.$$

The Right Ascension of the fictitious mean Sun, affected by aberration, and so taken as to have a uniform motion in the plane of the Earth's equator corresponding to the motion of the mean Sun in longitude at the epoch 1900;

$$\tau = 18^h\ 38^m\ 45^s.836 + 8\ 640\ 184^s.542\ T + 0^s.0929\ T^2$$

Notes

From above, we have

$$L = 18^h\ 38^m\ 47^s.2 + 8\ 640\ 184^s.542\ T + 0^s.0726\ T^2$$

Or

$$\tau = 279^\circ \frac{41459}{60000} = 279.69098333333335\dots^\circ + \dots$$

$$L - \tau = (279 + 41459/60000) - (279 + 62701/90000)$$

$$L - \tau = (41/7200)^\circ = 20.5'' \text{ (exactly).}$$

On page 12, Newcomb (1898) gives a value $20''.501 = 0.005\ 694\ 722 \dots^\circ$ for aberration.

The first terms differs from that in τ because of aberration. The second term matches. Why don't the third terms match?

From USNO 1961 page 22: “*The Besselian solar year*. For certain applications it is more convenient to measure time in units of tropical centuries of 3652.21988 ephemeris days, the fundamental epoch being the beginning of the Besselian (fictitious) solar year 1900.0, or 1900 January 0^d.813 E.T. In the great majority of such cases the difference in length of the century is not significant: the same symbol T is accordingly used, though always with a specific explanation. The difference between the lengths of the Besselian solar year and the tropical year ($0^s.148 T$) can always be neglected and multiples of 0.01 in T thus relate to the beginning of the corresponding Besselian year (see section 2B).”

Newcomb’s L vs τ — T^2 Discrepancy: Math Summary

Given formulas:

$$L = 279^\circ 41' 48.04'' + 129\,602\,768.13'' T + 1.089'' T^2$$

$$\tau = 18^h 38^m 45.836^s + 8\,640\,184.542^s T + 0.0929^s T^2$$

Step 1 — Convert τ ’s linear term to arcseconds (using $1^s = 15''$):

$$8\,640\,184.542^s \times 15 = 129\,602\,768.13''$$

$$\Rightarrow \tau_{T^1} \times 15 = L_{T^1} \quad \checkmark \text{ (exact match)}$$

Step 2 — Convert τ ’s T^2 term to arcseconds:

$$0.0929^s \times 15 = 1.3935''$$

Step 3 — Compare to L ’s T^2 term:

$$\Delta_{T^2} = 1.3935'' - 1.089'' = 0.3045''/\text{century}^2$$

Step 4 — Equivalently, convert L ’s T^2 term to time-seconds (dividing by 15) and compare directly in time units:

$$L_{T^2}^{(time)} = \frac{1.089''}{15} = 0.0726^s$$

$$\Delta_{T^2}^{(time)} = 0.0929^s - 0.0726^s = 0.0203^s/\text{century}^2$$

Step 5 — Cross-check via unit conversion:

$$0.0203^s \times 15 = 0.3045''$$

$$\Rightarrow \Delta_{T^2}^{(time)} \times 15 = \Delta_{T^2} \quad \checkmark \text{ (self-consistent)}$$

Conclusion:

$$\Delta_{T^2} = 0.3045''/\text{century}^2 = 0.0203^s/\text{century}^2$$

This is documented in the *Explanatory Supplement to the Ephemeris* as the “**excess of the secular acceleration of the right ascension of the fictitious mean sun over the mean longitude of the Sun,**” and manifests as the Besselian year being shorter than the tropical year by:

$$\delta_{year} \approx 0.148^s T$$

(a linear-in- T effect, consistent with being the time-derivative of a T^2 position offset, since $\frac{d}{dT}(k T^2) = 2kT$).

The beginning of the solar year 1900, known also as the BESSELIAN fictitious year ; defined as the moment when the Right Ascension of the fictitious mean sun is 280° ;

1900. January $0^d.3135$, Greenwich Mean Time.

The Earth's mean anomaly ;

$$g = 358^\circ 28' 33''.0 + 129\,596\,579.10'' T - 0''.54 T^2 - 0''.012 T^2$$

The eccentricity of the Earth's orbit;

$$e = 0.016\,751\,04 - .000\,041\,80T - .000\,000\,126 T^2$$

$$= 3455''.150 - 8''.621 T - 0''.0260 T^2$$

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The obliquity of the ecliptic;

$$\epsilon = 23^\circ 27' 8''.26 - 46''.845 T - 0''.0059 T^2 + 0''.001\,81 T^2$$

2 HMNO 1961

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3. Universal time

Universal time is the precise measure of time used as the basis for all civil time-keeping; it conforms with a very close approximation of the mean diurnal motion of the Sun. *

*See note on page vi regarding the current basis of civil time scales. In genral the term “universal time” (U.T.) may be identified throughout this Supplement with the system of U.T.I defined on page 86.

It is, and since the introduction of Newcomb's *Tables of the Sun* has been, defined as 12 hours + the Greenwich hour angle of a point on the equator whos right ascension, measured from the mean equinox of date, is: ...

$$R_U = 18^h 38^m 45^s.836 + 8640184^s.542 T_U + 0s.0929 T_U^2$$

where T_U is the number of Julian centuries of 36525 days of universal time elapsed since the epoch of Greenwich mean noon (regarded as 12^h U.T.) on 1900 January 0. The expression for R_U is identical with that given by Newcomb (*Tables of the Sun, A.P.A.E.*, 6, part I, page 9, 1895) for the right ascension of the fictitious mean sun, with the exception that Newcomb used T instead of T_U and did not specify in what measure of time T was to be reconed. Newcomb, not recognising the variable rotation of the Earth, considered that T was measured in mean solar time applicable alike to orbital motions and to hour angles; as explained in sub-section B.I, Newcomb's T may now be identified with ephemeris time. The poin on the equator whose right ascension is R_U is not identical with the “fictitious mean sun” as defined by Newcomb; the right ascension of the fictitious mean sun is:

$$R_U = 18^h 38^m 45^s.836 + 8640184^s.542 T_E + 0s.0929 T_E^2$$

where T_E is the number of Julian centuries of 36525 days of ephemeris time elapsed since the epoch of 12^h E.T. on 1900 January 0. R_E differs from R_U by $0.002738 \Delta T$, where ΔT is the difference E.T. – U.T.

The implications of this distinction are considered in sub-section B.4.

The measure of universal time at time T_U , expressed in hours, minutes, and seconds, is thus:

$12^h +$ the Greenwich hour angle of the mean equinox of date – R_U

The date expressed in the form either of a calendar date or of a Julian date (see sub-section B.I), is that corresponding to the time T_U .

...

Greenwich mean sidereal time = U.T. + R_U + 12^h

for 0^h U.T. of every day; at U.T. = 0^h the value of the right-hand side is obtained by adding 12^h to the expression R_U for the mean sidereal time of 12^h U.T., and the relation becomes

G.M.S.T of 0^h U.T. = 6^h 38^m 45^s.836 + 86 40184^s.542 T_U + 0^s.0929 T_U^2 where T_U takes on successive values at a uniform interval of 1/36525.

...

3 Russell 1971

Appendix 2. The Calculation of the Position of the Sun

G.D. Mead (private communication) has written a simple subroutine to calculate the position of the Sun in GEI coordinates. It is accurate for years 1901 through 2099, to within 0.006 deg. The input is the year, day of year and seconds of the day in UT. The output is Greenwich Mean Sidereal Time in degrees, the ecliptic longitude, apparent right ascension and declination of the Sun in degrees. The listing of this program follows. We note that the cartesian coordinates of the vector from the Earth to the Sun are:

```
X = cos(SRASN) cos(SDEC)
Y = sin(SRASN) cos(SDEC)
Z = sin(SDEC)
```

```

SUBROUTINE SUN(IYR, IDAY, SECS, GST, SLONG, SRASN, SDEC)      1
C PROGRAM TO CALCULATE SIDEREAL TIME AND POSITION OF THE SUN.  2
C GOOD FOR YEARS 1901 THROUGH 2099. ACCURACY 0.006 DEGREE.  3
C INPUT IS IYR, IDAY (INTEGERS), AND SECS, DEFINING UN. TIME.  4
C OUTPUT IS GREENWICH MEAN SIDEREAL TIME (GST) IN DEGREES,    5
C LONGITUDE ALONG ECLIPTIC (SLONG), AND APPARENT RIGHT ASCENSION  6
C AND DECLINATION (SRASN, SDEC) OF THE SUN, ALL IN DEGREES    7
DATA RAD /57.29578/                                           8
DOUBLE PRECISION DJ, FDAY                                     9
IF(IYR. LT. 1901. OR. IYR. GT. 2099) RETURN                  10
FDAY = SECS/86400                                             11
DJ = 365* (IYR-1900) + (IYR-1901)/4 + IDAY + FDAY -0.5D0    12
T = DJ / 36525                                                13
VL = DMOD (279.696678 + 0.9856473354*DJ, 360.D0)           14
GST = DMOD (279.690983 + 0.9856473354*DJ + 360.*FDAY + 180., 360.D0) 15
...                                                            16
```

Notes

The code above also appears in Geopack-2008.

On IYEAR = 1900, IDAY = 1, FDAY = 0.5, DJ = 0, so

$$\text{GST} = (279.690983 + 360.0 \times 0.5 + 180.0) \times 360.0$$

$$\text{GST} = (279.690983 + 360.0) \times 360.0$$

$$\text{GST} = 279.690983$$

Newcomb 1898 page 9 gives “the Right Ascension of the fictitious mean Sun, affected by aberration”, as

$$\tau = 18^{\text{h}} 38^{\text{m}} 45^{\text{s}}.836 + 8\,640\,184^{\text{s}}.542T + 0.0929T^2$$

where T is the number the number of Julian centuries of 36525 days since 1900, Jan 0, Greenwich Mean Noon. This can be written as

$$\begin{aligned}\tau &= 279.69098333333335\dots^\circ + (8640184^{\text{s}}.542) \frac{15^\circ}{3600^{\text{s}}} T + 0^{\text{s}}.0929T^2 \\ &= 279.69098333333335\dots^\circ + (36000.7689249999966705\dots^\circ) T + 0^{\text{s}}.0929T^2\end{aligned}$$

Using $T = \text{DJ}/36525$ gives

$$\tau = 279.69098333333335\dots^\circ + (0.9856473353867213\dots^\circ)\text{DJ} + 0^{\text{s}}.0929T^2$$

which matches

$$\text{GST} = 279.690983 + 0.9856473354 \times \text{DJ}$$

to the precision used in GST and if the T^2 is ignored.

When $\tau = 0$, corresponding to 1900, Jan. 0, Greenwich Mean Noon $\tau = 279.69098333333335\dots^\circ$, which matches the R71 value 279.690983 to the precision given even though the R71 equation is said to be valid for only $\text{year} > 1901$).

4 Hapgood 1992

Section 3.1

All of the fundamental transformations defined in the following sections are time dependent. To maintain a uniform style, time is there specified by modified Julian date (MJD), which is the time measured in days from 00:00 UT on 17 November 1858 (Julian date 2400000.5). In this paper we use only the integer part of MJD, i.e. the value at 00:00 UT on the day of interest. For some applications it is also necessary to give the time within the day as Universal Time in hours (UT).

Section 4.1

The rotation angle θ is the Greenwich mean sidereal time. This can be calculated using the following formula (U.S. Naval Observatory, 1989):

$$\theta = 100.461 + 36000.770 T_0 + 15.04107 \text{ UT},$$

where

$$(3) \quad T_0 = \frac{\text{MJD} - 51544.5}{36525.0}.$$

Note that T_0 is the time in Julian centuries (36525 days) from 12:00 UT on 1 January 2000 (known as epoch 2000.0) to the previous midnight.

Notes

1.

In Hapgood (1992) UT has two meanings: 1. Universal Time and 2. the fractional hour of the day. In the following equations UT should be read UT (similar to notation used in USNO 1989).

2.

[U.S. Naval Observatory, 1989](#) does not give the Hapgood (1992) equation for θ explicitly. It gives, on page B2, eqn 2:

$$\text{GMST} = 6^{\text{h}}.69737456 + 2400.051336 T_0 + 0.0000258622 T_0^2 + 1.002737909 UT$$

[note the missing dimensions on the last three coefficients] with definitions on page B3:

T_0 and T are time intervals in Julian centuries from J2000.0:

$$T_0 = (\text{JD}_0 - 2451545.0)/36525 \quad T = (\text{JD} - 2451545.0)/36525;$$

UT is the universal time in hours;

JD_0 and JD are the Julian dates at 0^{h} UT and at an arbitrary time of the day, respectively;

Converting to degrees using

15*6.69737456 = 100.4606184	1
15*2400.051336 = 36000.77004	2
15*0.0000258622 = 0.000387933	3
15*1.002737909 = 15.041068635	4
	5
	6
	7

gives

$$\text{GMST} = 100.4606184^{\circ} + (36000.77004^{\circ}) T_0 + 0.000387933^{\circ} T_0^2 + (15.041068635^{\circ}) UT$$

and Hapgood (1992) is

$$\theta = 100.461 + 36000.770 T_0 + 15.04107 UT$$

3.

θ in Hapgood (1992) follows from Eqn 12.3 for Θ_0 in [Meeus \(1998\)](#):

$$\Theta_0 = 100.46061837^{\circ} + 36000.770053608 T + 0.000387933 T^2 - T^3/38710000$$

with $T = (\text{JD} - 2451\,545.0)/36525$. This equation in Meeus (1998) is followed by the statement “... To obtain the sidereal time θ_0 at Greenwich for any instant UT of a given date, multiply that instant by 1.002 737 909 35 and add the result to the sidereal time Θ_0 at 0^{h} UT.”

Doing so gives

$$\theta_0 = 100.46061837^{\circ} + 36000.770053608 T + 0.000387933 T^2 - T^3/38710000 + (15.0410686402499998...) UT$$

where $15 \cdot 1.00273790935 = 15.0410686402499998...$ was used. Eqn 3 for θ in Hapgood (1992) has the same constants as θ_0 rounded to the nearest 0.001 degree and the T^2 and T^3 terms omitted.

5 Meeus (1998)

Chapter 12

We shall denote by Θ_0 the sidereal time at Greenwich at 0^h UT of a given date, and by θ_0 the sidereal time at Greenwich for any given instant UT.

The sidereal time at the meridian of Greenwich, at 0^h Universal Time of a given date, can be obtained as follows.

Calculate the JD corresponding to that date at 0^h UT (Chapter 7). Thus, this is a number ending on .5. Then find T by

$$(12.1) \quad T = \frac{\text{JD} - 2451\,545.0}{36525}$$

The *mean* sidereal time at Greenwich at 0^h UT is then given by the following expression which was adopted in 1982 by the International Astronomical Union:

$$(12.2) \quad \Theta_0 = 6^{\text{h}}41^{\text{m}}50^{\text{s}}.54841 + 8640\,184^{\text{s}}.812\,866\,T + 0^{\text{s}}.093\,104\,T^2 - 0^{\text{s}}.000\,0062\,T^3$$

Expressed in degrees and decimals, this formula can be written

$$(12.3) \quad \Theta_0 = 100.460\,618\,37 + 36000.770\,053\,608\,T + 0.000\,387\,933\,T^2 - T^3/38\,710\,000$$

Important: the formulae (12.2) and (12.3) are valid only for those values of T which correspond to 0^h UT of a date. All other values would give incorrect results.

To obtain the sidereal time θ_0 at Greenwich for any instant UT of a given date, multiply that instant by 1.002 737 909 35 and add the result to the sidereal time Θ_0 at 0^h UT.

The mean sidereal time at Greenwich, expressed in *degrees*, can also be found directly for any instant as follows. If JD is the Julian Day corresponding to that instant in UT (not necessarily 0^h), find T by formula (12.1), and then

$$(12.4) \quad \theta_0 = 280.460\,618\,37 + 360.985\,647\,366\,29 (\text{JD} - 2451\,545.0) + 0.000\,387\,933\,T^2 - T^3/38710000$$

Notes

From the notes for Hapgood (1992), we can write

$$\theta_0 = 100.46061837^\circ + 36000.770053608\,T + 0.000387933\,T^2 - T^3/38710000 + (15.0410686402499998\dots)\text{ UT}$$

6 Hapgood 1995

Section 3

The Russell (1971) paper is more complex. GST and $\lambda_\odot$ are given by the Fortran subroutine in his Appendix 2. The statements in that routine are functionally equivalent to the equations in Hapgood (1992) and thus yield values in the mean epoch-of-date. ...

However, the values of GST and $\lambda_\odot$ predicted by Russell agree with those of Hapgood (1992) to within 0.01°.

7 Fränz and Harper (2002)

Section 1

We also cite the formulae and methods given by Hapgood (1992) for geocentric systems, which are based on the Astronomical Almanac for Computers (1988) which is no longer updated by the Nautical Almanac Offices. The formulae used by Hapgood (1992) are first order approximations of the third order formulae given in Expl. Suppl. (1961).

Section 2

The Julian Day Number (JD) starts at Greenwich mean noon 4713 Jan. 1, B.C. [S. 2.26]. The *epoch day number* is defined in this paper as the fractional number of days of 86400 seconds from the epoch:

$$(1) \quad d_0 = (JD - 2451545.0).$$

Formulae from S. and A. use Julian centuries (T_0) from J2000.0. One Julian century has 36525 days, one Julian year has 365.25 days, s.t. [S. T3.222.2]

$$(2) \quad T_0 = d_0/36525.0 \quad \text{and} \quad y_0 = d_0/365.25.$$

Section 3.3.1

For the precision needed in this paper, we may neglect the difference between T_U and T_0 , such that (Meeus, 2000):

$$(20) \quad \text{GMST} \approx 280^\circ.46061837 + 360^\circ.98564736629 d_0 + 0^\circ.0003875 T_0^2 - 2^\circ.6 \cdot 10^{-8} T_0^3.$$

Notes

$1/38710000 (= 2.583311805...10^{-8})$ is given in Meeus (1998) Eqn 12.4 and $2.6e - 08$ is used in Fränz and Harper (2002) above (and Siedelman 1992). Also, 0.000387933 is used in Meeus (1998) Eqn 12.4 (and Siedelman 1992) but 0.0003875 is used in Fränz and Harper (2002). Also, $0.000387933 T^2$ used in Meeus (1998) Eqn 12.4 (and Seidelmann 1992) but $0.0003875 T_0^2$ is used in Fränz and Harper (2002).

Finally, Fränz and Harper 2002 cite “Meeus, J, 2000, Astronomical Algorithms 2nd Edition” but but I am only able to find Meeus, J, 1998, Astronomical Algorithms 2nd Edition.